

AULA004 - atividade

1 - Estação 1

De caracteres a lexemas e tokens

```
PS C:\Users\Unicesumar> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp'
Main'
ALUNO: Wellington Amorim de Sousa | RA: 24214926-2
ENTRADA: total_WA = valor + 26;

LEXEMA          TOKEN
-----
total_WA        IDENTIFICADOR
=               ATRIBUICAO
valor           IDENTIFICADOR
+              SOMA
26              INTEIRO
;              PONTO_E_VIRGULA
PS C:\Users\Unicesumar> |
```

01_tokens.png

2 - Estação 2

O que o analisador léxico reconhece

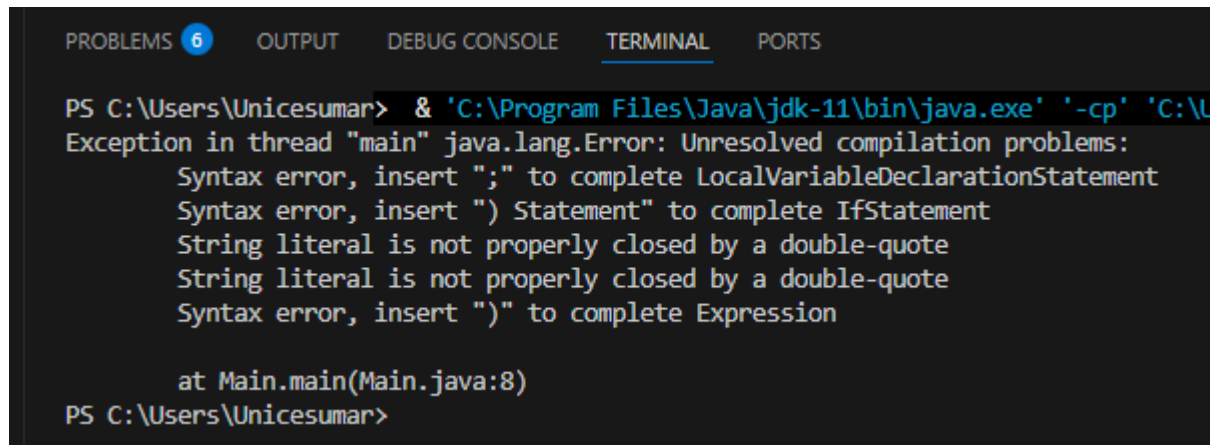
```
=== TESTE C2 - identificador personalizado ===
ENTRADA: int Wallet = 10;
LEXEMA          CLASSE          TOKEN
-----
int             LETRA           PALAVRA_RESERVADA
Wallet          LETRA           IDENTIFICADOR
=              SIMBOLO         ATRIBUICAO
10             DIGITO          INTEIRO
;              SIMBOLO         PONTO_E_VIRGULA

=== TESTE D - caractere inválido ===
ENTRADA: int valor# = 1;
LEXEMA          CLASSE          TOKEN
-----
int             LETRA           PALAVRA_RESERVADA
valor          LETRA           IDENTIFICADOR
#              DESCONHECIDO     ERRO_LEXICO
=              SIMBOLO         ATRIBUICAO
1              DIGITO          INTEIRO
;              SIMBOLO         PONTO_E_VIRGULA
PS C:\Users\Unicesumar> |
```

02_lexico.png

3 - Estação 3

Quando o compilador encontra erros

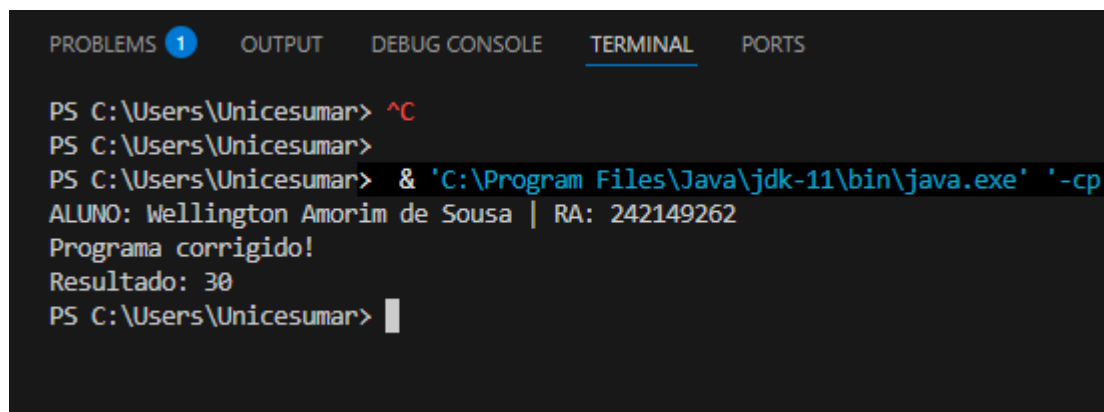


A screenshot of an IDE's terminal window. The terminal has tabs for PROBLEMS (with a blue circle containing the number 6), OUTPUT, DEBUG CONSOLE, TERMINAL (which is selected), and PORTS. The terminal text shows a command to run a Java program, followed by an exception message: "Exception in thread 'main' java.lang.Error: Unresolved compilation problems:". Below this, five specific error messages are listed: "Syntax error, insert ';' to complete LocalVariableDeclarationStatement", "Syntax error, insert ')' Statement' to complete IfStatement", "String literal is not properly closed by a double-quote", "String literal is not properly closed by a double-quote", and "Syntax error, insert ')' to complete Expression". The stack trace indicates the error occurred at "Main.main(Main.java:8)". The prompt "PS C:\Users\Unicesumar>" is visible at the bottom.

```
PS C:\Users\Unicesumar> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp' 'C:\U
Exception in thread "main" java.lang.Error: Unresolved compilation problems:
    Syntax error, insert ";" to complete LocalVariableDeclarationStatement
    Syntax error, insert ") Statement" to complete IfStatement
    String literal is not properly closed by a double-quote
    String literal is not properly closed by a double-quote
    Syntax error, insert ")" to complete Expression

    at Main.main(Main.java:8)
PS C:\Users\Unicesumar>
```

03_erro_compilador.png



A screenshot of an IDE's terminal window. The terminal has tabs for PROBLEMS (with a blue circle containing the number 1), OUTPUT, DEBUG CONSOLE, TERMINAL (which is selected), and PORTS. The terminal text shows a command to run a Java program, followed by the output: "ALUNO: Wellington Amorim de Sousa | RA: 242149262", "Programa corrigido!", and "Resultado: 30". The prompt "PS C:\Users\Unicesumar>" is visible at the bottom.

```
PS C:\Users\Unicesumar> ^C
PS C:\Users\Unicesumar>
PS C:\Users\Unicesumar> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp'
ALUNO: Wellington Amorim de Sousa | RA: 242149262
Programa corrigido!
Resultado: 30
PS C:\Users\Unicesumar> █
```

04_programa_corrigido.png

4 - Estação 4

Parser descendente recursivo

```

PS C:\Users\Unicesumar> & 'C:\Program Files\Java\jdk-11\bin\java.exe'
Main'
ALUNO: Wellington Amorim de Sousa | RA: 242149262
ENTRADA: (2 + 3) * 4

ENTRANDO em <expr> | lookahead = (
  ENTRANDO em <term> | lookahead = (
    ENTRANDO em <factor> | lookahead = (
      Abrindo parenteses
      Novo lookahead: 2
      ENTRANDO em <expr> | lookahead = 2
        ENTRANDO em <term> | lookahead = 2
          ENTRANDO em <factor> | lookahead = 2
            Numero reconhecido: 2
            Novo lookahead: +
            SAINDO de <factor> | lookahead = +
          SAINDO de <term> | lookahead = +
        Encontrado operador +
        Novo lookahead: 3
      ENTRANDO em <term> | lookahead = 3
        ENTRANDO em <factor> | lookahead = 3
          Numero reconhecido: 3
          Novo lookahead: )
        SAINDO de <factor> | lookahead = )
      SAINDO de <term> | lookahead = )
    SAINDO de <expr> | lookahead = )
    Novo lookahead: *
    Fechando parenteses
  SAINDO de <factor> | lookahead = *
  Encontrado operador *
  Novo lookahead: 4
  ENTRANDO em <factor> | lookahead = 4
    Numero reconhecido: 4
    Novo lookahead: $FIM$
  SAINDO de <factor> | lookahead = $FIM$
  SAINDO de <term> | lookahead = $FIM$
  SAINDO de <expr> | lookahead = $FIM$

ENTRADA ACEITA
RESULTADO DA EXPRESSAO: 20

```

05_descendente.png

5 - Estação 5

Parser ascendente: deslocar e reduzir

```

PS C:\Users\Unicesumar> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp' 'C:\User
Main'
ALUNO: Wellington Amorim de Sousa | RA: 242149262
ENTRADA: id+id*id

PILHA          ENTRADA          ACAO
-----
$ 0            id+id*id$        SHIFT id -> estado 5
$ 0 id 5       +id*id$        REDUCE F -> id
$ 0 F 3        +id*id$        REDUCE T -> F
$ 0 T 2        +id*id$        REDUCE E -> T
$ 0 E 1        +id*id$        SHIFT + -> estado 6
$ 0 E 1 + 6    id*id$         SHIFT id -> estado 5
$ 0 E 1 + 6 id 5 *id$         REDUCE F -> id
$ 0 E 1 + 6 F 3 *id$         REDUCE T -> F
$ 0 E 1 + 6 T 9 *id$         SHIFT * -> estado 7
$ 0 E 1 + 6 T 9 * 7 id 5 id$   SHIFT id -> estado 5
$ 0 E 1 + 6 T 9 * 7 F 10 $    REDUCE F -> id
$ 0 E 1 + 6 T 9 $            REDUCE T -> T * F
$ 0 E 1 + 6 T 9 $            REDUCE E -> E + T
$ 0 E 1        $            ACCEPT
PS C:\Users\Unicesumar>

```

06_ascendente.png